

MISUN KIM

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RESEARCH INTERESTS

I study the reliability, safety, and controllability of interventions in foundation models, focusing on when similar internal representations in multilingual LLMs lead to different behavioural responses under cross-lingual and cross-task transfer.

EDUCATION

MSc Speech and Natural Language Processing Sep 2025 - Sep 2026

University of Sheffield | Sheffield, UK

- Predicted: Distinction
- Dissertation: Cross-lingual generalisation of representation-level interventions in multilingual LLMs

BE in Industrial Management Engineering Mar 2019 - Feb 2024

Hankuk University of Foreign Studies | South Korea

- Double Major: Central Asian Studies (Kazakh language and regional studies) | GPA 3.94/4.5
- First Distinction Scholarship (2023)

RESEARCH EXPERIENCE

MSc Dissertation Research — Multilingual LLMs Feb 2026 - Present

University of Sheffield | Supervisor: Dr Cass Zhao

- Investigated the robustness of activation-level interventions in multilingual LLMs using Contrastive Activation Addition (CAA), testing whether English-derived steering vectors transfer consistently across three model families and five languages.
- Found that high cross-lingual representational similarity did not reliably predict behavioural transfer, revealing a gap between geometric alignment and intervention robustness.
- Designed layer-wise evaluations with bootstrap confidence intervals, held-out sets, and Gaussian and permutation random-vector baselines to characterise failure modes and distinguish systematic effects from noise.
- Extended the analysis to TruthfulQA and cross-task settings, finding that interventions targeting one behaviour can also shift performance on another task and motivating mechanistic analysis of off-target effects.

Undergraduate Researcher Mar 2023 - Nov 2023

ICT Mentoring Project, Ministry of Oceans and Fisheries | Supervisor: Gyu-Young Lee (KAIST)

- First-authored LSTM study on container freight forecasting across five universities, reducing RMSE by 71% over the baseline (published ACK 2023)

KEY PROJECTS

Multilingual Post-Training Safety Evaluation Framework Aug 2026 - Present

- Built a reproducible multilingual LLM safety-evaluation pipeline for SFT, DPO, and RLVR checkpoints, combining typed experiment/result schemas, benchmark-native scoring, deterministic parallel-sample validation, and automated GitHub Actions CI.
- Ran a 50-item parallel EN/KO pilot on OLMo 3 7B using A40 GPUs, finding a consistent harmful-request refusal gap between Korean and English across all three post-training stages, motivating larger-scale evaluation of cross-lingual safety consistency.

ESG Compliance Checker

Mar 2026 - Present

- Built an end-to-end compliance verification system over 11 corporate ESG reports, mapping regulatory criteria to retrieved evidence and producing traceable covered/partial/missing judgements using hybrid retrieval and LLM-assisted verification.
- Developed and audited a manually annotated retrieval gold set; hybrid retrieval with reranking improved Hit@5 from 83.3% to 100% and MRR from 0.628 to 0.819 on a manually annotated retrieval gold set.
- Used retrieval and verifier error analysis to identify annotation gaps and evidence-selection failures, then modularised the system with FastAPI, automated testing, Docker, and a separate Streamlit interface.

SKILLS

Programming: Python, SQL

ML & NLP: PyTorch, Hugging Face Transformers, scikit-learn, clustering (k-means, hierarchical)

LLM Interpretability & Safety: activation steering (CAA), representation analysis, layer-wise interventions, cross-lingual behavioural evaluation, post-training safety evaluation, refusal analysis, experimental controls, LLM-as-judge

Research Infrastructure: HPC/SLURM, Weights & Biases, Git, Docker, FastAPI, Flask, ChromaDB

PUBLICATIONS & MANUSCRIPTS

- Kim, M.*, Kim, Y., Kim, E., Lee, B., Han, Y., & Lee, G. (2023). Implementation of Container Volume Prediction Technology using Deep Learning. ACK Conference Proceedings, 30(2).
- Manuscript in preparation (2026). Cross-lingual and cross-task generalisation of activation-level interventions in multilingual LLMs.

WORK EXPERIENCE

Sustainability Research Analyst

Mar 2025 - May 2025

Eon consulting group | South Korea

- Contributed to 170+ page sustainability reports for major public and financial organisations, conducting TNFD climate scenario analysis and developing a CSDDD-aligned supply-chain risk assessment framework.

Field Service Manager (Data & Operations)

Payhere | South Korea

Mar 2024 - Aug 2024

- Built workflow automation using the Slack API, Google Sheets, and Looker Studio to analyse 10,000+ service records, reducing unnecessary field visits by 17%.

TEACHING EXPERIENCE

Language Tutor (Private)

Mar 2021 - Nov 2025

- Provided individualized Korean/English instruction to 5+ learners aged 6–18, from early literacy to CSAT-level advanced reading

LANGUAGES

Korean (Native), English (Fluent), Spanish (Intermediate), Kazakh (Intermediate)